

Shahab Derhami

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Education

Ph.D. Industrial & Systems Engineering, Auburn University, AL, 2017 (GPA: 4.0/4.0)
M.E. Industrial & Systems Engineering, Auburn University, AL, 2012 (GPA: 4.0/4.0)
M.S. Social & Economic Systems Engineering, Bu-Ali Sina University, Iran, 2006
B.S. Industrial Engineering, Azad University, South Tehran branch, Iran, 2004

Employment

Senior Research Scientist, Georgia Institute of Technology, GA, 2018-Present
Postdoctoral Fellow, Georgia Institute of Technology, GA, 2017-2018
Instructor, Research Assistant, and Teaching Assistant, Auburn University, AL, 2011-2017
Economic Analyst, PartLastic Factories Group, Iran, 2008-2011
Production & Project Planning Specialist, Atlas Afrooz Shargh Company, Iran, 2006-2008
Instructor, Department of Industrial Engineering, Azad University, Iran, 2004-2005

Publications

Journal papers

1. **S. Derhami**, J. S. Smith, and K. R. Gue, [Space-efficient layouts for block stacking warehouses](#), *IIE Transactions*, DOI:10.1080/24725854.2018.1539280.
2. **S. Derhami**, J. S. Smith, and K. R. Gue, [Optimising space utilisation in block stacking warehouses](#), *International Journal of Production Research*, vol. 55, no. 21, pp. 6436–6452, 2017.
3. **S. Derhami** and A. E. Smith, [An integer programming approach for fuzzy rule-based classification systems](#), *European Journal of Operational Research*, vol. 256, no. 3, pp. 924–934, 2017.
4. **S. Derhami** and A. E. Smith, [A technical note on the paper “hGA: Hybrid genetic algorithm in fuzzy rule-based classification systems for high-dimensional problems”](#), *Applied Soft Computing*, vol. 41, pp. 91–93, 2016.

Refereed conference proceedings

1. B. Montreuil, S. Buckley, L. Faugère, R. Khir, and **S. Derhami**, [Urban parcel logistics hub and network design: The impact of modularity and hyperconnectivity](#), *Progress in Material Handling Research: 2018*.
2. **S. Derhami**, J. S. Smith, and K. R. Gue, [A simulation model to evaluate the layout for block stacking warehouses](#), *Progress in Material Handling Research: 2016*.
3. **S. Derhami** and A. E. Smith, [Iterative mixed integer programming model for fuzzy rule-based classification systems](#), in *2014 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*, 2014, pp. 2079–2084.

Under review manuscripts

1. **S. Derhami** and B. Montreuil, Estimation of potential lost sales in retailers of high-value substitutable products, *Under Review*.

In-preparation manuscripts

1. Optimizing space utilization and transportation costs in block stacking warehouses: A simulation approach, with Jeffrey S. Smith and Kevin R. Gue.
2. Optimal assortment planning for retailers of high-value substitutable products, with Benoit Montreuil and Iman Dayarian.
3. Optimal assortment planning under inventory transshipment policies, with Iman Dayarian and Benoit Montreuil.
4. Assessing product availability in the retail distribution networks with shared inventory, with Benoit Montreuil and Guilhem Bau.
5. Maximizing product availability under customer substitution and stochastic demand, with Jinyong Yim and Benoit Montreuil and Iman Dayarian.

Teaching experience

Instructor

- Auburn University (Fall 2015)
Stochastic Operations Research (98 students, avg. review: 5.0/6.0)
- Business Trading University, Iran (2007)
Introduction to Manufacturing Processes, Production Management
- Azad University, Iran (2004-2005)
Plant Layout, Inventory Management, Production Planning, Application of Computer in IE

Teaching assistant

Operations Planning & Control, Stochastic Operation Research, Simulation (graduate), Probability & Statistics I, Simulation Modeling & Analysis, Deterministic Operations Research

Honors & awards

- Finalist in the INFORMS data mining best student paper for the paper “An integer programming approach for fuzzy rule-based classification systems”, 2016
- Honorable award for the poster “A novel optimization approach for data mining: a medical diagnosis case”, Graduate Engineering Research Showcase, Auburn University, 2015 (\$1,000)
- Conveyor & Sortation Systems Honor Scholarship, the Material Handling Education Foundation, Inc., 2015 (\$2,000)
- Departmental award for the poster “Optimizing Space Utilization in Block Stacking Warehouses”, Graduate Engineering Research Showcase, Auburn University, 2014 (\$500)
- Storage equipment manufacturers association honor scholarship, the Material Handling Education Foundation, Inc., 2012 (\$3,000)
- Auburn University graduate travel grants to attend three conferences, 2014-2016 (\$2,800)
- Graduate teaching assistantship, Auburn University, 2011-2017
- Graduate teaching assistantship, Bu-Ali Sina University, 2005-2006

Conference presentations and seminars

- “A simulation-based approach to estimate demand across retail networks and supply chains of high-value products”, IISE Annual Meeting, Orlando, FL, 2018
- “A novel optimization approach for disease diagnosis, poster presentation”, 4th Annual Postdoctoral Symposium at Georgia Institute of Technology, Atlanta, GA, 2017

- “Designing space-efficient warehouse layouts”, INFORMS Annual Meeting, Houston, TX, 2017
- “An integer programming approach for fuzzy rule-based classification systems”, Best data mining student paper competition, INFORMS Annual Meeting, Nashville, TN, 2016
- “Designing layouts for block stacking warehouses”, poster presentation, INFORMS Annual Meeting, Nashville, TN, 2016
- “Optimal design for block stacking warehouses”, INFORMS Annual Meeting, Nashville, TN, 2016
- “Optimizing space utilization in block stacking warehouses”, INFORMS Annual Meeting, Philadelphia, PA, 2015
- “Operations research in data mining: an application to medical diagnosis”, INFORMS Annual Meeting, Philadelphia, PA, 2015
- “Optimizing space utilization in designing block stacking warehouses”, IIE Annual Meeting, Nashville, TN, 2015
- “Ant colony optimization for work load smoothing in assembly lines”, 6th International Industrial Engineering Conference, Tehran, Iran, 2009
- “Using ant colony algorithm to solve multi objective assembly line balancing problem”, 2nd International Operation Research Conference, Babolsar, Iran, 2009
- “Solving fuzzy assembly line balancing problem with ant algorithm”, 3rd Joint Congress on Fuzzy and Intelligent Systems, Yazd, Iran, 2009

Service to profession

- Reviewer for European Journal of Operational Research, International Journal of Production Research, Computers & Operations Research, IEEE Transactions on Automation Science and Engineering, Applied Mathematical Modelling, SIMULATION: Transactions of The Society for Modeling and Simulation International, and 2015 IIE Annual Conference
- Session chair in 2018 INFORMS Annual Meeting
- Session chair in 2018 IIE Annual Meeting
- Session chair and student volunteer in 2015 IIE Annual Conference
- Member of Institute for Operations Research and the Management Sciences (INFORMS)
- Member of Institute of Industrial & Systems Engineers (IIE)